

HOW TO MAKE THE 3D DESIGN USING AUTODESK INVENTOR

Thursday, May 22, 2025 9:43 AM

Step 1

Install the software. During this step, one could install either Autodesk Inventor/ Solid Works/ AutoCAD for CAD design. <https://www.autodesk.com/education/edu-software/overview#INVPROSA>

Select the student version in order to get a cheaper version.

Step 2.

Navigate to the File of results and Identify which ring you will design first. For example, the first ring is the center ring, 842. The inner bore is 120mm, Outer bore is 190mm. Then the outer-most circle where the shim trays are added is 240mm.

Step 3.

After looking at design parameters, open Autodesk Inventor. Select File> New> Templates> Metric> (Standard)mm> Create. This sets the dimensions from the default (Inches to mm).

The display of the background is usually dark so one can also change this to sky by selecting (File> Options> Colors> Sky.

Then select apply to apply these changes.

Step 4

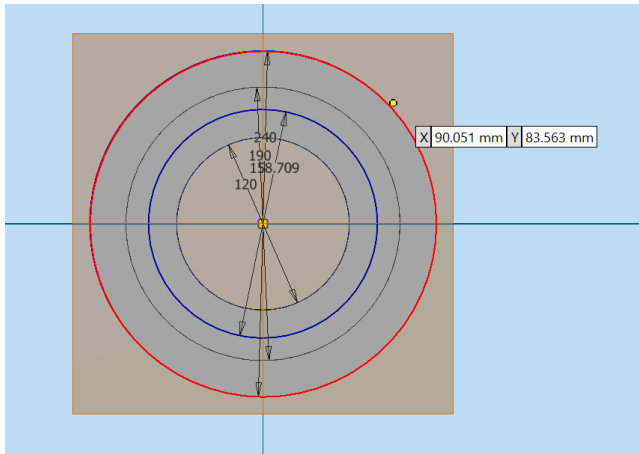
When the new part is created, all the planes are invisible. To make them visible, select Origin> Right click Planes> Visible. Do this for the YZ, XZ and XY planes.

Right click on the XY plane and make it perpendicular (or navigate to the Sketch menu> select the perpendicular sign).

Step 5: Draw circles

To draw the inner bore, navigate to the Sketch > Center circle. Draw the circle while starting at the center. Set the diameter to 120mm. Draw a second circle for the outer bore of the ring and set its diameter as 190mm. Draw a third circle (for the shim trays) and set its diameter to 240mm.

For this case, Inner bore diameter = 80mm, Outer bore diameter = 160mm and the outermost bore diameter = 190mm.

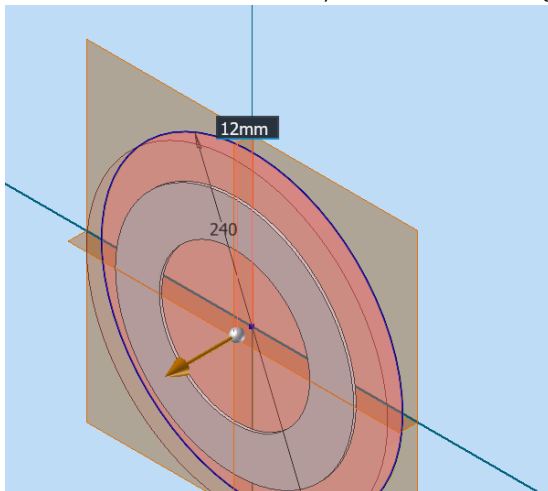


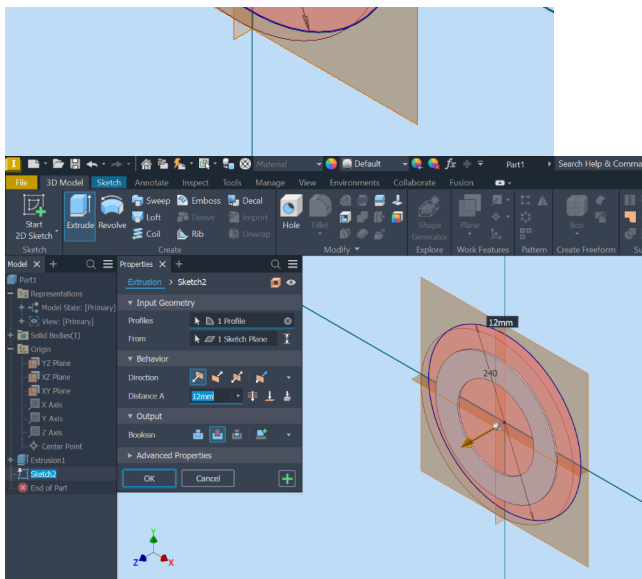
Step 6: Extrude the circles

After drawing the circles, finish sketch. First extrude the inner circle by 14mm. Finish the sketch.

Then right click on the sketch and select visible. Then extrude the second face by 12mm.

Select the inner section and extrude it by 14mm. Select the outer region and extrude it by 12mm.



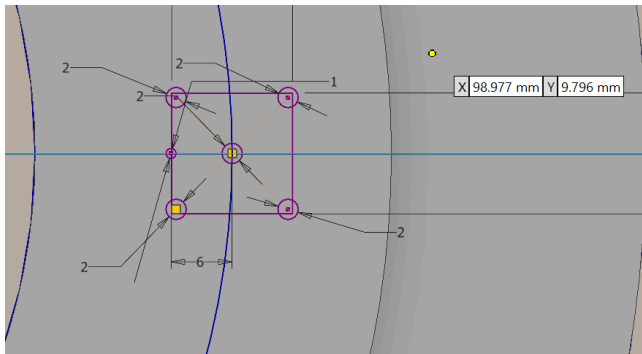


Step 7: Drawing the magnet Band

From the field results, band-radius is shown. For example, the 842 middle ring as a band radius of 78.702mm. Draw a circle. Set the entire diameter as 157.404mm. Press enter. That is the band where the magnets will be.

Step 8: Create Magnet Block

Select the Sketch 5, where the band-radius line is and select edit sketch to draw the magnet block. Select the two side rectangle and draw at the horizontal line that crosses the radius band circle at 180 degrees.



The dimensions of the magnet are 11.8mmX 11.8mmX 11.8mm after measuring. So dimension the rectangle as 11.8mm. So it forms a square.

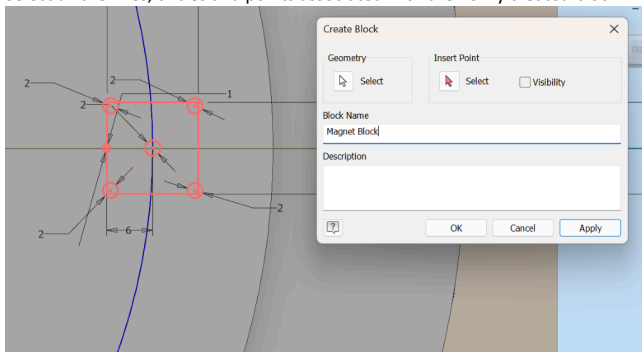
Draw a little dot on the side of the square facing the midpoint of the entire ring. This will give the direction in which the magnets will rotate. Dimension it such that the distance between the centre of the dot and center of the square is 6mm. Set the diameter of the dot to 1mm.

Draw other small circles at the four corners of the square. Set their diameters to 2mm each. There are such that the magnet can be fit into the hole with ease.

Draw another circle at the center of the square and set its radius to 2mm. This is so as to ease the process of removing a magnet if necessary.

To create a magnet block navigate to **Folder Model> Block> Create Block/ Sketch>Edit Sketch**

Select all the lines, circles and points associated with the newly created block.

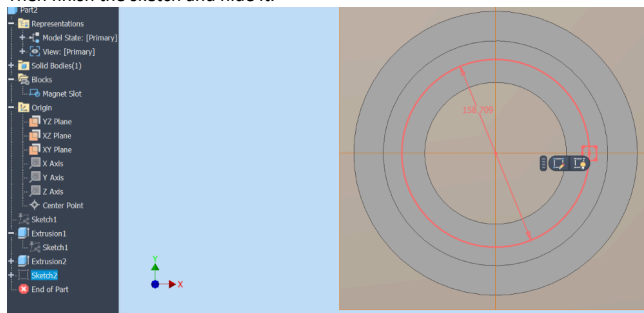


Select insert point and choose the center of the block. This selects the point at which the block will be inserted at each point of the band-radius. Input a block Name as Magnet Block . Click on apply to create it. All the center points of circles and rectangle should change to color white.

Click the green tick to finish.

You can hide the other sketches to create more focus on the magnet block.

Then finish the sketch and hide it.



Still within the sketch mode, draw a new radii band circle and make its diameter 157.404. Right click on the radii band circle, select edit block. Then right click on the magnet block, select place block and place it at the first point.

Place a magnet block onto it.

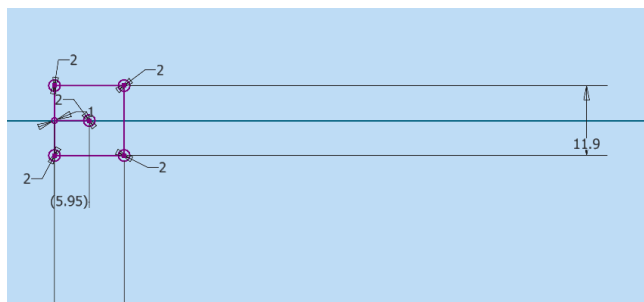
If you try moving the block, it will move entirely. In order to fix it, draw a line from the center of the ring to its corresponding point on the radii band circle. Now place the magnet block at the point where this horizontal line meets the band. The center of the block should turn green.

Try to move the block around and it should rotate about the point at 360 degrees

From the results file, identify the number of magnets in the selected band for that specific ring. For example, the center ring 842 has 20 magnets.

Here, a pattern is made of the magnet blocks. Select Edit Sketch (the radii band circle with the magnet block), Choose the circular pattern option then for axis, select the center of the ring and for geometry select the horizontal line from the center of ring to center of block. [select the horizontal line from the center of the ring to the center of the magnet block. Select the ring center as the axis and make 20 patterns]. Select OK.

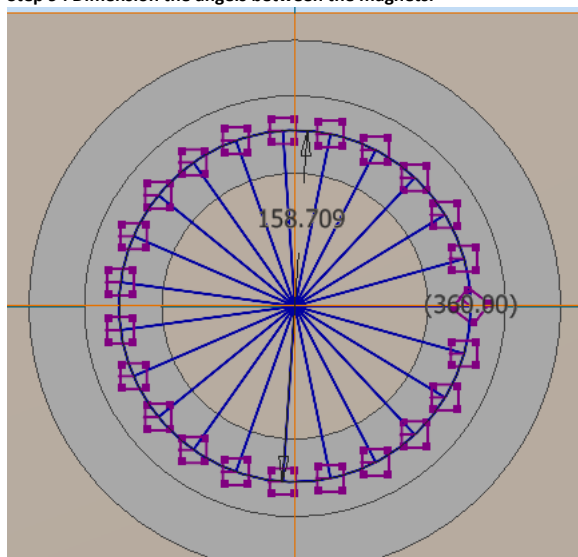
Select the magnet block from the menu and place it at every end of the line. Make sure the center point of the magnet (Insert point) turns green when the magnet block is placed at every point.

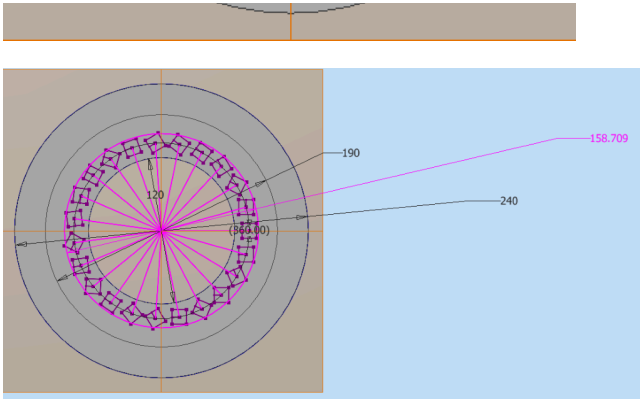


Then, still in the sketch mode, select edit block.

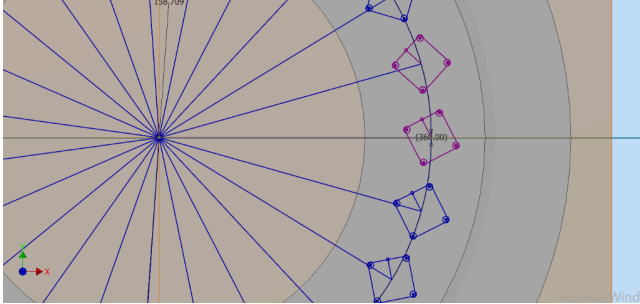
Draw a line from the center of the block to the center of the small circle placed 6mm away from the center of the magnet slot.

Step 9 : Dimension the angels between the magnets.





In preparation for dimensioning the magnet slots, tilt them so that they do not auto-connect to anything. This is done to make sure the dimensions are angular and not linear.



Step 10: DIMENSIONS FOR THE MAGNET SLOTS

Note: To simplify the dimensioning process, make sure that all the magnet slots are purple in color hence flexible. That way, one can change their dimensions and angle of orientation. If the magnet slots are blue, then they are already fixed.

Here use the formula for halbach array which is: $360(2)/n$. N is the number of magnets.

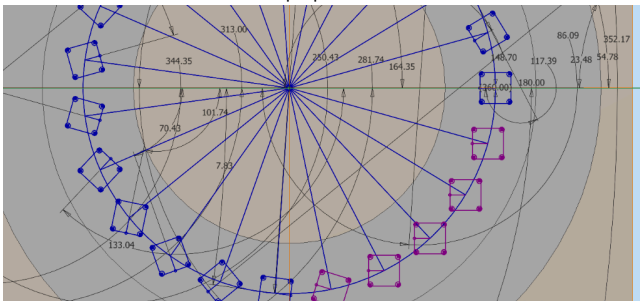
For example, for the 120mm inner bore diameter ring, the first ring to be designed is the middle ring 842. From the results obtained from the algorithm, the ring has 20 magnets .

$360(2)/20 = 36$.

This means there is an angular difference of 36 degrees for each slot.

However, this value is a negative because the magnet slots are rotated in the anti-clockwise direction

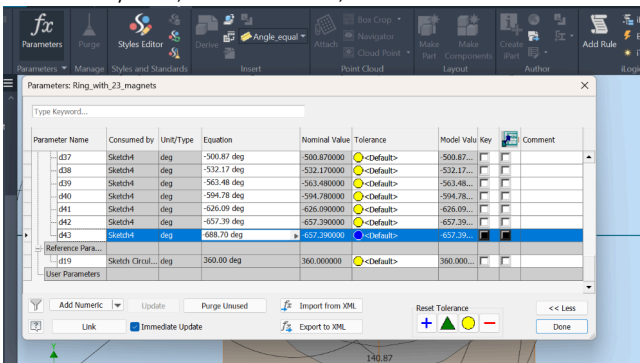
After setting the specific angular dimension for a slot, it attains a fixed orientation. The fixed slots are blue while the flexible ones are purple.



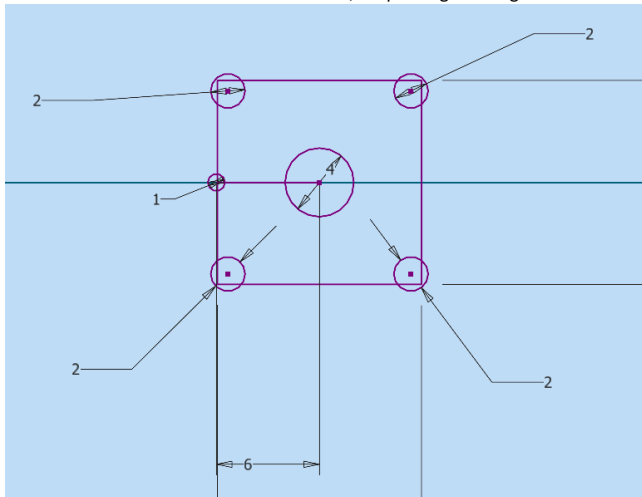
After putting all the angular dimensions, all the rings should appear blue.

Next, navigate to **Manage > Parameters**. This enables one to look through all the angles and ensure that they are the right ones.

If there are any errors, correct them. Then, save.

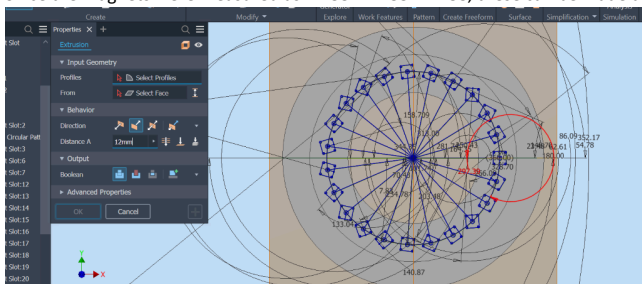


Edit the block and add a circle at the center, for pushing out magnets



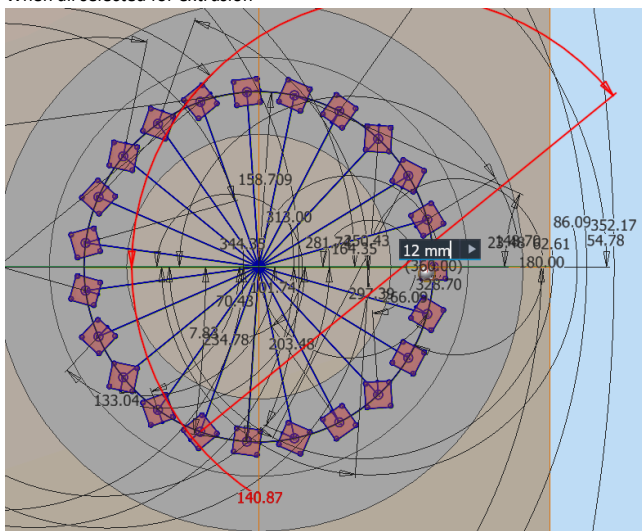
STEP: EXTRUDE THE MAGNET SLOTS.

Since the magnets were measured as 11.88X 11.88X 11.88, these can be made as 12mm.

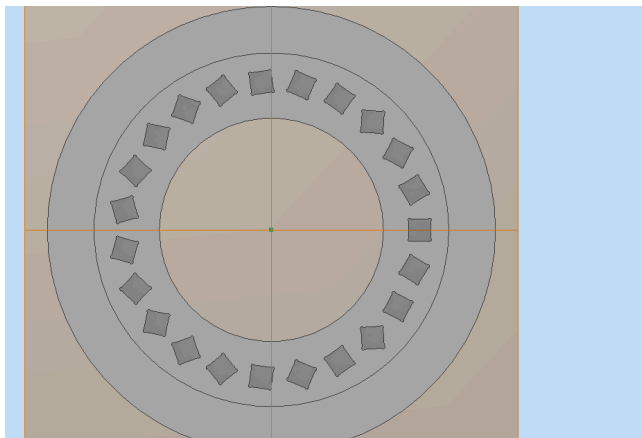


For this extrusion click once on each magnet slot, and the four pressure relief circles at each corner.

When all selected for extrusion

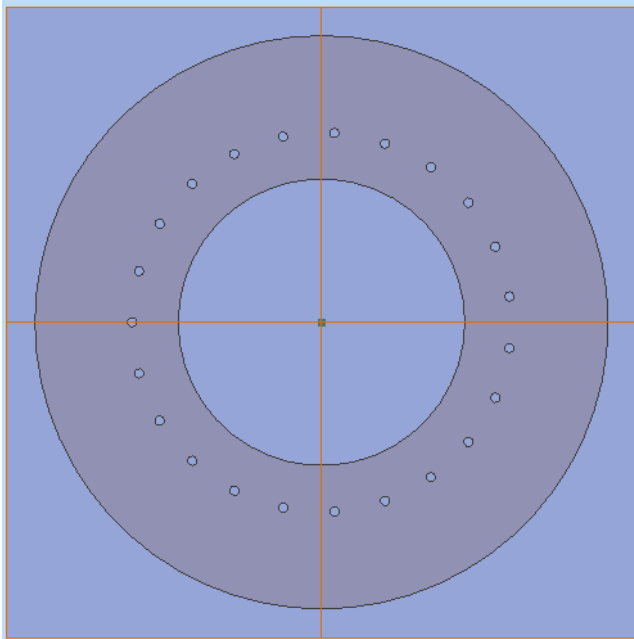


After extrusion





Now extrude cut the holes.



First, make the circles drawn for the hole extrusion visible.

Then, select extrude.

Later, draw the small circles for inserting the screws and lids.

After this action, draw the M4 circles where the brass rods will pass.